



Tag 13
08.07.2026
mittwoch

Von KI erstellt, von Menschen kontrolliert.

Guten Morgen!

GitHub Repo <https://github.com/www42/a5>

Morgen: Mannheim

AZ-104

Learning Paths:

- Manage identities and governance in Azure
- Prerequisites for Azure administrators – CloudShell and templates
- Configure and manage virtual networks for Azure administrators
- Implement and manage storage in Azure
- Deploy and manage Azure compute resources VM, Docker, Kubernetes, Function, Logic App
- Monitor and back up Azure resources

Heute

- SAS , Rekeying (MeasureUp)
- Mini-Übung Key Azure for Student ✓
- Skillable Challenge Lab

-
- Start LP Compute
 - Create VM Portal Azure for Student
Delete

What happened yesterday



SAS

Create Shared Access Signatures

- Provides delegated access to resources
- Grants access to clients without sharing your storage account keys
- The account SAS delegates access to resources in one or more of the storage services
- The service SAS delegates access to a resource in just one of the storage services

Was ist eine Signatur:

Eine Signatur ist ein verschlüsselter Hash

video.mpl

SHA256
~~7f7b280815...~~
versch. on

on key

(= Prüfsumme)

MD5, SHA1, SHA2, SHA256...

Signing method ^①
☒ Account key ☐ User delegation key

Signing key ^①
Key 1

Permissions ^①
Read

Start and expiry date/time ^①
Start
02/01/2021
(UTC-08:00) Coordinated Universal Time-08

Expiry
02/02/2021
(UTC-08:00) Coordinated Universal Time-08

Allowed IP addresses ^①
for example, 168.1.5.65 or 168.1.5.65-168.1....

Allowed protocols ^①
☒ HTTPS ☐ HTTP

Generate SAS token and URL

SHA256

2x7abc
on key 1

Mini-Übung

List and Rotate Storage Access Keys (Azure CLI)

Azure for Student

→ GitHub a5

→ Cloud Shell

→ Local Shell (VS Code) *

Schweizer

Erich Gamma

Eclipse

In this challenge, you will configure network peering between two Azure virtual networks. First, you will review the existing environment. Next, you will configure Azure virtual network peering. Finally, you will create two virtual machines by using Azure PowerShell, and then you will use the virtual machines to test the virtual network peering connections. Note: Once you begin the challenge, you will not be able to pause, save, or return to your challenge. Please ensure that you have set aside enough time to complete the challenge...

Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

3



[Implement an Azure Load Balancer \[Guided\]](#)

Challenge Labs All Access Pass , AZ300.2-007

⌚ 30 minutes

In this Challenge Lab, you will implement an Azure load balancer. First, you will create an Azure load balancer. Next, you will configure a backend pool for the load balancer, and then you will configure an HTTP health probe. Finally, you will create a load balancing rule. Note: Once you begin the challenge, you will not be able to pause, save, or return to your challenge. Please ensure that you have set aside enough time to complete the challenge before you start.

Additional Details



Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

8

Challenge Labs supporting AZ-104: Implement and manage storage in Azure

1



[Manage Access to Azure Storage \[Guided\]](#)

Challenge Labs All Access Pass , AZ300.1-002

⌚ 30 minutes

In this Challenge Lab, you will secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using...

Additional Details



Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

9

Challenge Labs supporting AZ-104: Deploy and manage Azure compute resources

1



[Enable Azure Virtual Machine Scale Sets for High Availability and Scalability \[Guided\]](#)

Challenge Labs All Access Pass , HCS-009

⌚ 30 minutes

In this challenge, you will create and deploy multiple Azure virtual machine scale sets acting as web server and app server tiers. First, you will create a virtual machine scale set for the web server tier. Next, you will use an Azure Resource Manager (ARM) template to create a virtual machine scale set for the app server tier. Finally, you will verify connectivity to the virtual machine scale set for the web server tier. Note: Once you begin a challenge you will not be able to pause, save, or return to your progress. Please ensure you have set...

Additional Details



Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes
Allow Multiple Attempts: Yes (Limit 5 attempts)

10

Challenge Labs supporting AZ-104: Monitor and back up Azure resources

1



[Implement Azure Backup for Azure Virtual Machines \[Guided\]](#)

Challenge Labs All Access Pass , AZ300.2-009

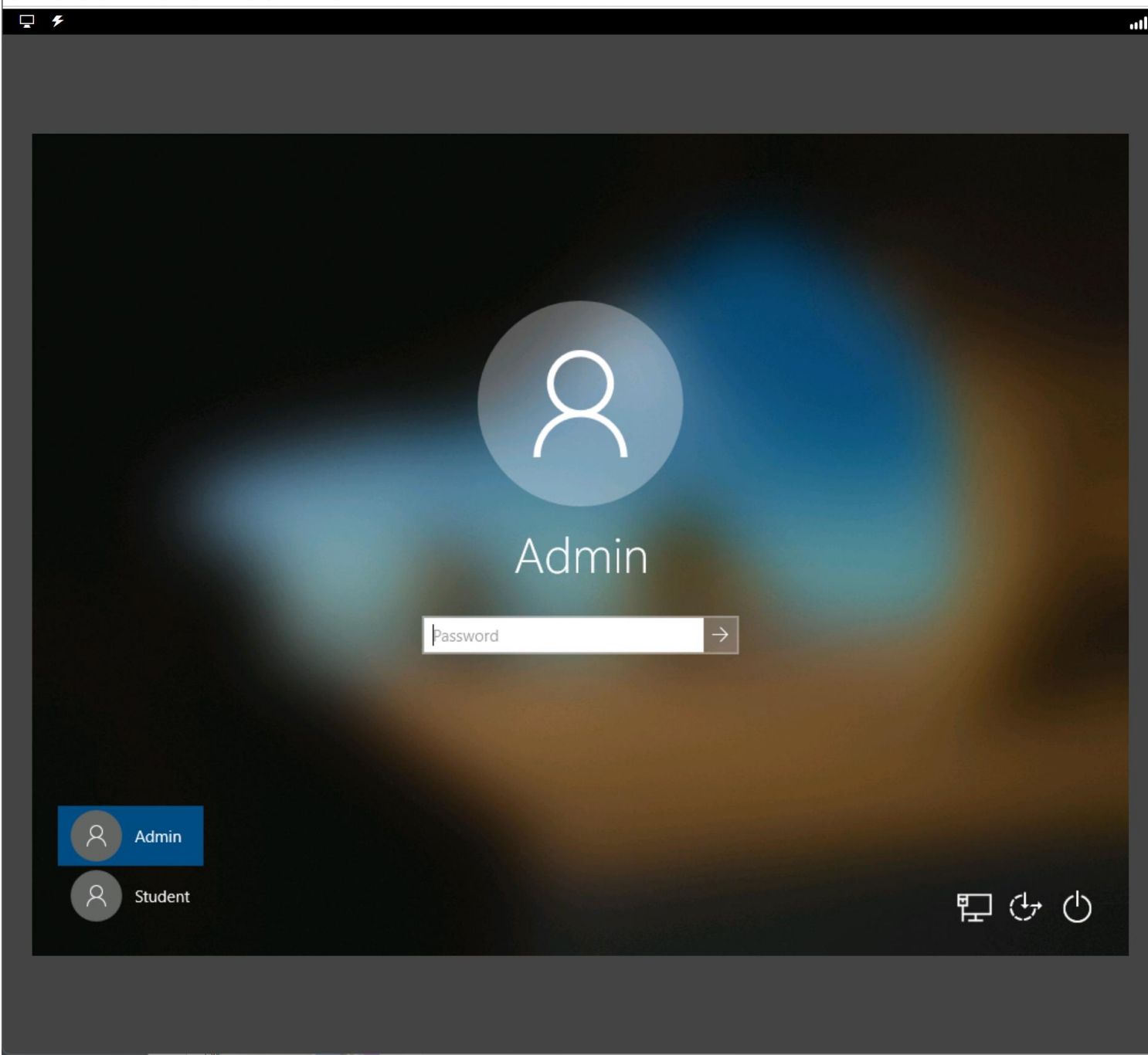
⌚ 30 minutes

In this challenge, you will implement Azure backup for your Azure virtual machines. First, you will back up an Azure virtual machine by using the Azure portal. Next, you will enable

Additional Details



Required: No
Available Instructor-Led: Yes
Available Self-Paced: Yes



AZ300.1-002: Manage Access to Azure Storage [Guided]
30 Minutes Remaining

InstructionsResourcesHelp

skillable challenges

Guided

Manage Access to Azure Storage

Challenge Overview

Understand the scenario

You are an Architect with Hexelo and responsible for your organization's Azure environment. You need to secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using Microsoft Azure Storage Explorer.

Navigating the Challenge Lab

Understanding a Cloud Slice

Quick tips for navigating the Challenge Lab instructions.

Next: Configure a storage account to... >

The screenshot shows the Windows 10 Settings application with the 'Language' settings page open. The left sidebar shows 'Time & Language' selected. The main pane displays the 'Language' settings. A red circle highlights the 'Deutsch (Deutschland)' entry, which is the default app language and has the language pack installed. Below it, 'English (United States)' is listed as the Windows display language. A red arrow points from the 'Deutsch (Deutschland)' entry to the taskbar, which shows the language code 'DEU'.

AZ300.1-002: Manage Access to Azure Storage [Guided]

23 Minutes Remaining

Instructions Resources Help

Guided

Manage Access to Azure Storage

Challenge Overview

Understand the scenario

You are an Architect with Hexelo and responsible for your organization's Azure environment. You need to secure an Azure storage account. First, you will configure a storage account to use a customer-managed storage service encryption key, and then you will configure the storage account to use secure transfer. Next, you will configure firewall rules to restrict storage account access to a specific IP address, and then you will generate a shared access signature to be used to grant time-limited, delegated access to a storage account. Finally, you will test the access to the storage account by using Microsoft Azure Storage Explorer.

Navigating the Challenge Lab

- Understanding a Cloud Slice
- Quick tips for navigating the Challenge Lab instructions.

Next: Configure a storage account to...

AZ-104

 Deploy and manage
Azure compute resources



VM

Vorteile

gibts schon lange

VMware

Microsoft: Hyper-V

Kapselung

Introduction to Azure Virtual Machines

Scalierbarkeit

VM Scale Set (VMSS)

Nachteile

Kosten (in Azure)

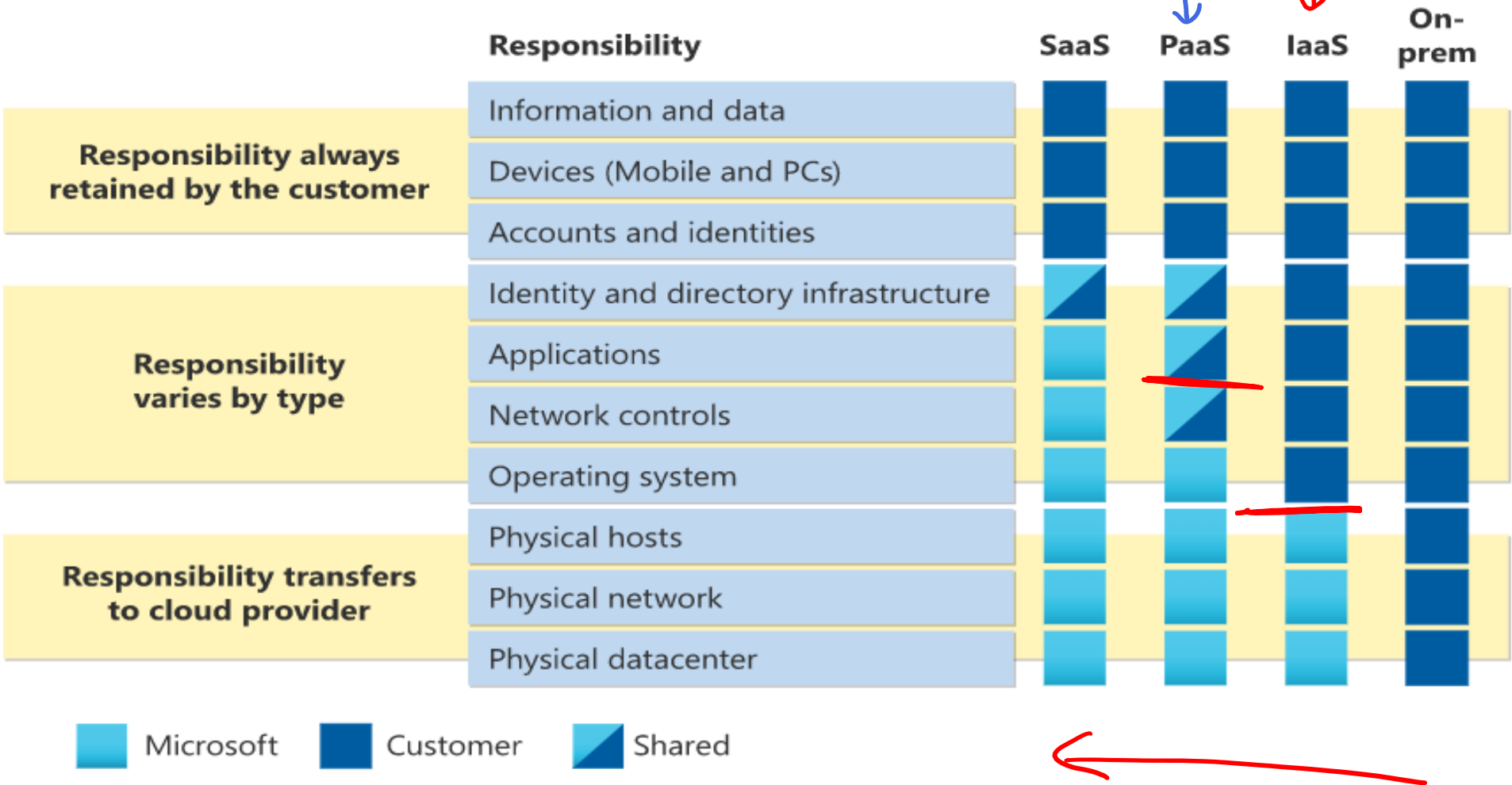
Ressourcen intensiv

Lange Boot
Zeiten

Snowflake Server

Docker
Container

Review Cloud Services Responsibilities



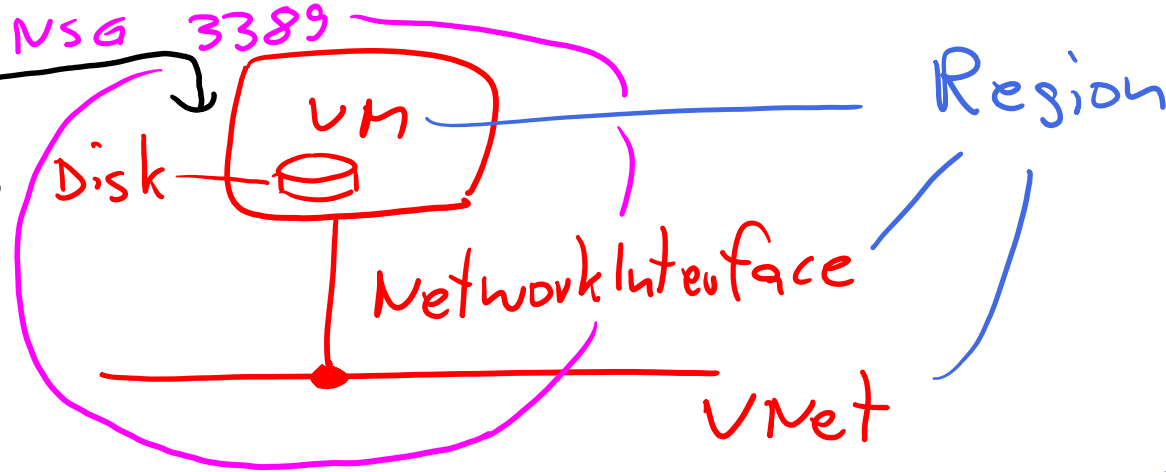
Plan Virtual Machines

- Start with the network
- Name the virtual machine
- Choose a location = Region
 - Each region has different hardware and service capabilities
 - Locate Virtual Machines as close as possible to your users and to ensure compliance and legal obligations
- Consider pricing

Compute Size
+ Disk

Standard-DS2v3

Besser: Bastion Host



Azure for Students
5 Regions



Determine Virtual Machine Sizing

<i>Family</i> Type	Description
General purpose B2	Balanced CPU-to-memory ratio.
Compute optimized	High CPU-to-memory ratio.
→ Memory optimized	High memory-to-CPU ratio.
Storage optimized	High disk throughput and I/O.
→ GPU	Specialized virtual machines targeted for heavy graphic rendering and video editing.
High performance compute	Our fastest and most powerful CPU virtual machines

(Lab 03 ? 5 Disks)

Determine Virtual Machine Storage

managed disk

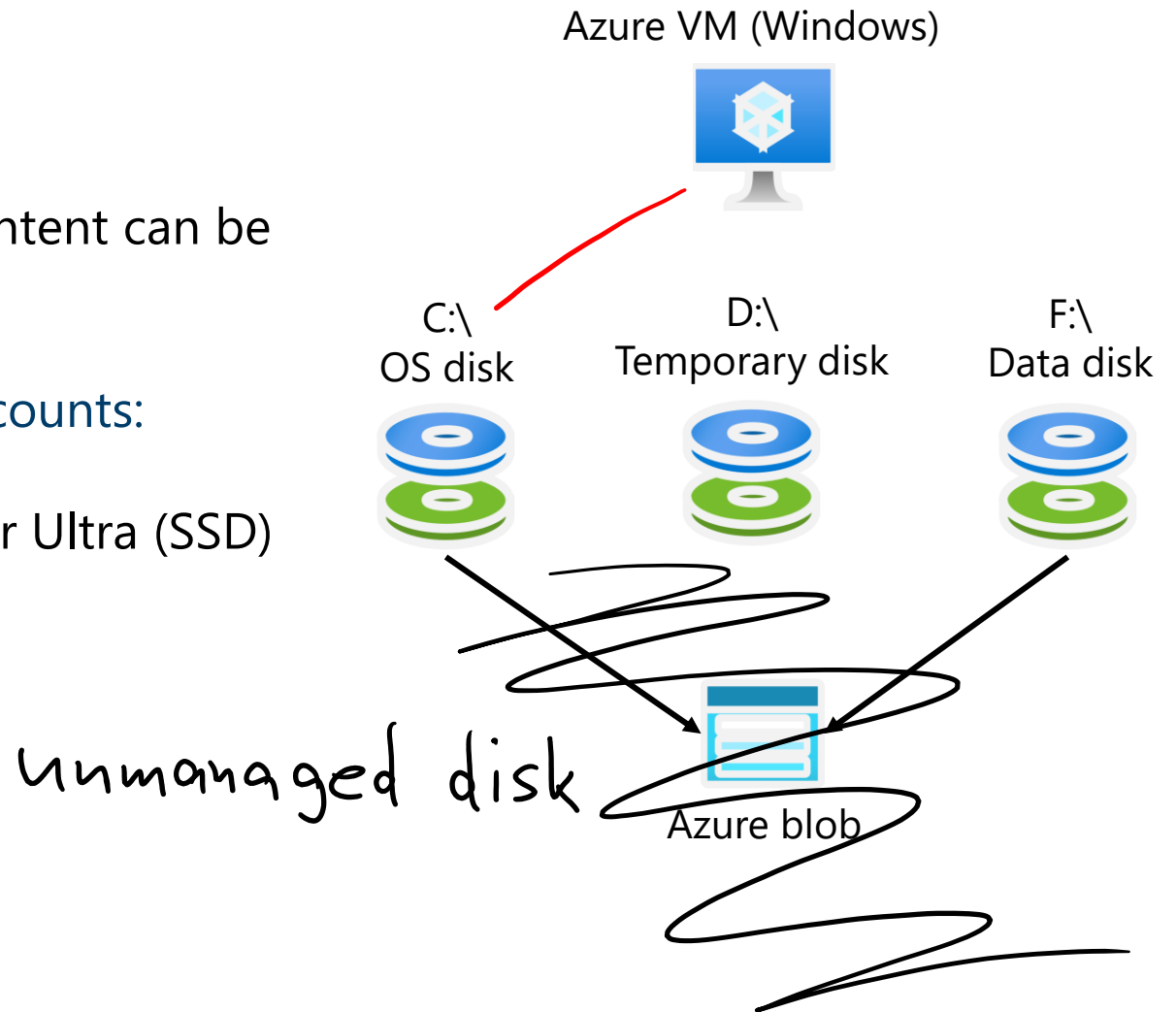
Each Azure VM has two or more disks:

- OS disk
- Temporary disk (not all SKUs have one, content can be lost)
- Data disks (optional)

OS and data disks reside in Azure Storage accounts:

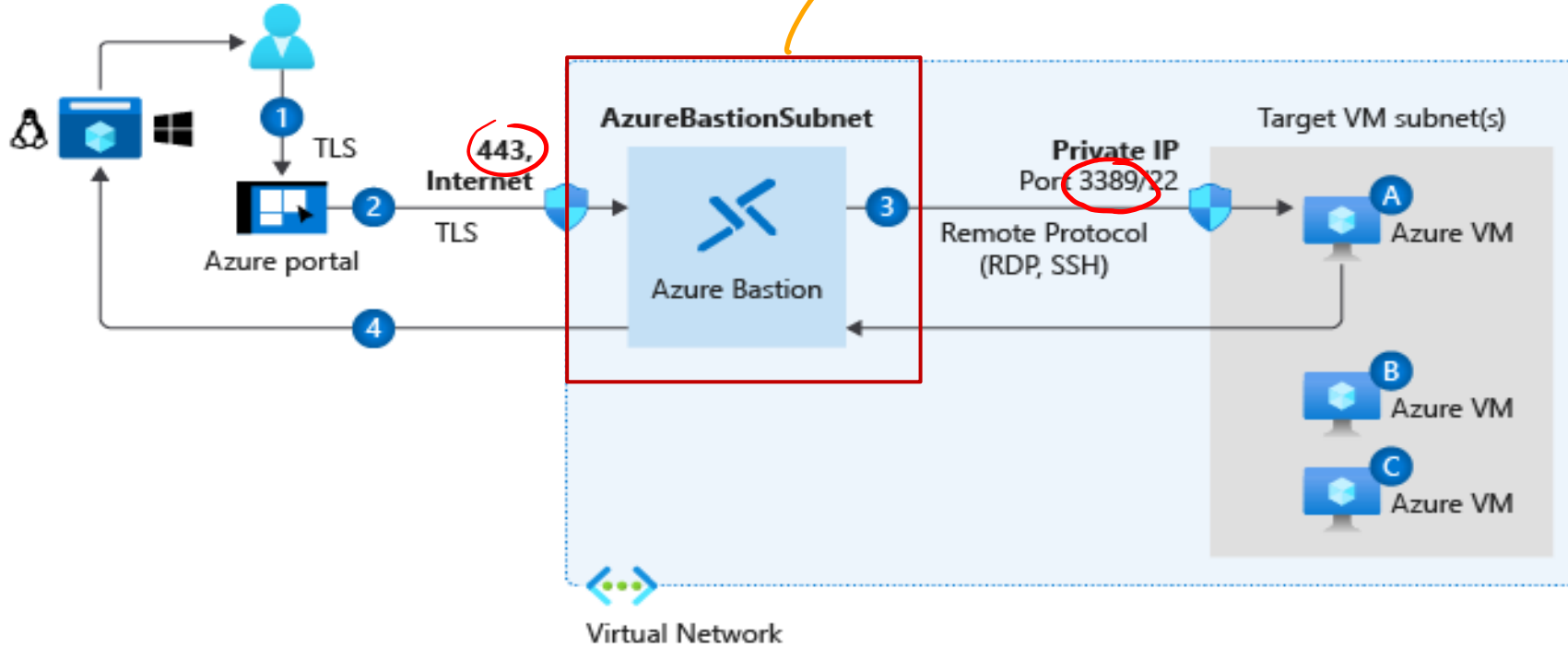
- Azure-based storage service
- Standard (HDD, SSD) or Premium (SSD), or Ultra (SSD)

Azure VMs use managed disks



Connect to Virtual Machines

Student SKU (Free)
(Developer?)



Bastion Subnet for RDP/SSH through the Portal over SSL

Remote Desktop Protocol for Windows-based Virtual Machines

Secure Shell Protocol for Linux based Virtual Machines

Labs



TO THE LAB ZONE →

NO THROUGH ROUTE
Berlin
Europa
Dublin

SPEEDY ROUTE

4²

lounge
Microsoft
tech-ed
2010

lounge

software diagnostics

E 94

SD Studio
for Team Foundation Server
Visualize your code in
a software map.

SD Developer Edit
for C and C++ for Visual
Trace dynamically your code
Visualize the control flow.

software
diagnostics

DataCore
SOFTWARE

Azure for Students

1) VM Deployment

Region
Windows
Size ? B2S...
ARM ? x86!
VNet
NSG

2) Connect to VM

Bastion Host (Free Tier)

3) Create Web Site
wenn Windows → IIS

4) Advanced Settings

5) Alles löschen ⚡

End of presentation

